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Decade–Long Timing Study of the two GMRT discovered Millisecond Pulsar J0248+4230 and J1207−5050

Sapan Kumar Sahoo [1], Bhaswati Bhattacharyya [1], and Jayanta Roy [1]

[1] National Centre for Radio Astrophysics, Tata Institute of Fundamental Research, S. P. Pune University Campus, Pune 411007, India

ABSTRACT

We present updated timing solutions for the isolated millisecond pulsars (MSPs) J0248+4230 and J1207−5050, both discovered with the Giant Metrewave Radio Telescope (GMRT). Using beyond decadal–long high–precision timing data from the upgraded GMRT (uGMRT), we refine their spin (P), astrometric, and dispersion measure (DM) parameters to unprecedented accuracy, achieving post–fit residuals of only a few microseconds. For the first time, we measure their proper motions, corresponding to transverse velocities of 15.3 $\mathrm{km\,s^{-1}}$ for J0248+4230 and 43.9 $\mathrm{km\,s^{-1}}$ for J1207−5050, consistent with the recycled MSP population. In addition, we detect unusually strong linear polarization ($\sim 93.25\%$) in J1207−5050–much higher than typically observed in MSPs–pointing to highly ordered magnetospheric emission, while in J0248+4230 we report the first detection of individual and bright pulses, offering new insight into emission physics. When compared with PTA-listed MSPs, both pulsars fall squarely within the key parameter space of P, DM, timing stability, and sky coverage. Particularly, their large angular separation with PTA–listed pulsars makes them particularly valuable for strengthening correlations in the Hellings & Downs curve, thereby enhancing PTA sensitivity. Together, these updated measurements and new findings, with significantly reduced parameter uncertainties, not only represent a substantial improvement over earlier studies but also firmly establish J0248+4230 and J1207−5050 as stable, well-characterized MSPs, adding two promising candidates to the expanding pool of pulsars vital for neutron star physics, emission studies, and gravitational–wave detection.

1. INTRODUCTION

Pulsars are highly magnetized, rotating neutron stars that emit beams of electromagnetic radiation along their magnetic axes. Formed during the core–collapse supernovae of massive stars, these compact objects–typically only about 20 kilometers in diameter–exhibit some of the most extreme physical conditions observed in the Universe (Hewish et al. 1968). When the pulsar's rotation periodically brings its beam into the line of sight of Earth, it is detected as a periodic radio pulse, allowing these objects to function as extraordinarily precise cosmic clocks (Lorimer & Kramer 2004). Their stable rotational behavior makes pulsars invaluable probes of relativistic physics, dense matter, and the structure of our Galaxy.

Among the pulsar population, millisecond pulsars (MSPs) are distinguished by their rapid rotation rates–generally below 30 milliseconds–and exceptional timing stability. These pulsars are believed to be "recycled" neutron stars that have been spun up through prolonged accretion of angular momentum from a binary companion during a low–mass X–ray binary (LMXB) phase (Alpar et al. 1982; Bhattacharya 1991). This evolutionary pathway not only accelerates their spin but also stabilizes their magnetic fields, allowing for remarkably clean and stable pulse profiles. They have very high spinning stability($\dot{P}$ 10^{-16} to 10^{-21} $\frac{s}{s}$), which is comparable to stability of atomic clocks (Lorimer & Kramer 2004). As a result, MSPs are foundational to precision astrophysics and are used in diverse applications ranging from testing gravitational theories to detecting nanohertz gravitational waves.

Pulsar timing is the cornerstone method for studying pulsar parameter variations with time. By measuring the pulse times of arrival (ToAs) at radio observatories over months to decades, astronomers can construct detailed timing models that describe a pulsar's rotation, spatial motion, and–for binary pulsars–their orbital dynamics (Lorimer & Kramer 2004). These models allow the derivation of critical parameters such as spin period derivatives ($\dot{P}$), position (Ra and Dec), proper motion (μ_T), parallax, and binary orbital elements. Over time, deviations from these models can reveal new physics, including the presence of companions, gravitational perturbations, or even phenomena like pulsar glitches and timing noise. Furthermore, changes in the dispersion measure (DM) over time help trace the interstellar medium's electron content along the line of sight (Hobbs et al. 2006a).

While many MSPs reside in binary systems, isolated MSPs–those without observable companions–offer a unique and complementary perspective. Their timing models are typically simpler, free from the complexities of binary orbital dynamics such as binary parameters (e.g., orbital period, eccentricity, and projected semi–major axis). This simplicity allows for higher precision in the determination of spin period (P) and astrometric parameters, which include the pulsar's position (RA, Dec), proper motion (μ_T), and parallax. Such precision makes isolated MSPs ideal targets for long–term stability studies. For example, PSR J2124–3358 and PSR J0030+0451 are well–known isolated MSPs with exceptionally stable and precise timing solutions, providing valuable benchmarks for rotational stability studies and contributions to pulsar timing arrays (Hobbs et al. 2010; Antoniadis et al. 2023; Lorimer & Kramer 2004). Investigating these isolated systems also offers insights into the evolutionary processes that disrupt or ablate binary companions, leading to isolated configurations. Moreover, the absence of orbital noise enhances their utility in pulsar timing arrays.

A significant frontier in pulsar astronomy is the detection of gravitational waves in the nanohertz regime using pulsar timing arrays (PTAs). PTAs function as a Galactic–scale detector, leveraging the consistent timing of a network of MSPs to identify correlated perturbations in TOAs induced by passing gravitational waves (Hobbs et al. 2010). The dominant expected signal is a stochastic background generated by an ensemble of supermassive black hole binaries throughout cosmic history. Recent breakthroughs, such as the reported evidence of a common–spectrum process by NANOGrav and other PTA collaborations (Agazie et al. 2023; Antoniadis et al. 2023), underscore the urgency and importance of incorporating new high–quality MSPs into PTA datasets. Isolated, stable MSPs with long timing baselines, such as the ones studied in this work, can substantially improve array sensitivity and angular sky coverage.

GMRT(Swarup 1991), located in Pune, India, has played a pivotal role in pulsar discovery and their timing studies. Comprising thirty 45–meter antennas spread across a 25–kilometer region, the GMRT is among the most sensitive low–frequency radio telescopes in the world. In addition to wide–field surveys, GMRT has been instrumental in targeted searches based on high–energy observations. PSRs J0248+4230 and J1207−5050 were identified as promising pulsar candidates from unassociated *Fermi*–LAT gamma–ray sources. As part of a coordinated effort with the Fermi Pulsar Search Consortium, legacy GMRT observations at 322 and 607 MHz toward unassociated Fermi–LAT gamma–ray sources led to the discovery of three millisecond pulsars (MSPs): PSR J0248+4230, PSR J1207−5050, and PSR J1536−4948. These discoveries, first reported by Bhattacharyya et al. (2021), have since been followed by more than a decade of dedicated timing observations with GMRT.

Among these, PSR J0248+4230 is an isolated MSP with a spin period of 2.60 ms and a dispersion measure (DM) of approximately 48.26 pc $^{-3}$. It exhibits a steep radio spectrum across GMRT bands, with flux densities of $\sim$7.5 mJy at 322 MHz and $\sim$0.9 mJy at 607 MHz. PSR J1207−5050 is also an isolated MSP, spinning at 4.84 ms with a DM of roughly 50.60 pc cm^{-3}. It has been detected only at 607 MHz with a flux density of $\sim$0.5 mJy; its non–detection at 322 MHz suggests a possible spectral turnover. Both pulsars were subsequently confirmed as gamma–ray emitters through pulsation detections with the *Fermi*-LAT, reinforcing their value as powerful probes of pulsar emission physics. These findings were first presented in the discovery study by Bhattacharyya et al. (2021), which also reported follow-up timing campaigns spanning approximately five years. These observations enabled phase-coherent timing solutions for both sources, allowing for measurements of their spin, astrometric parameters.

In this work, we present updated timing solutions for J0248+4230 and J1207−5050 using more than ten years of timing datasets. Section 2 describes the observations and data analysis methods, while Section 3 presents the improved timing models and residuals. In Section 4, we explore complementary gamma-ray properties, polarization results, and assess detections in optical surveys. In section 5, we study the polarization properties of PSR J1207−5050. Section 6 compares these two pulsars' parameter space with existing PTA–listed pulsar populations, and Section 7 reports the single–pulse and bright–pulse analyses. Finally, in Section 7 we discuss the broader implications of these results and summarize our conclusions.

2. OBSERVATIONS AND DATA ANALYSIS

Initial timing data (pre–2017) were acquired using the GMRT Software Backend (GSB) with the legacy GMRT systems in band–3 (centered at 322 MHz with a 33 MHz bandwidth) for PSR J0248+4230 and in band–4 (centered at 607 MHz with a 33 MHz bandwidth) for PSR J1207−5050. These early observations enabled the construction of their initial phase-connected timing solutions. Each session typically lasted 30–45 minutes.

After the discovery and localization of the MSPs–once their precise right ascension and declination were determined–they were observed using the Phased Array (PA) mode of the GMRT. This mode coherently combines signals from N antennas and offers a sensitivity improvement by a factor of $\sqrt{N}$ compared to the Incoherent Array (IA) mode. The enhanced sensitivity allowed for higher signal-to-noise ratio (S/N) detections and improved timing precision. Subsequent follow-up observations were conducted using the upgraded GMRT (uGMRT, Gupta et al. 2017) after 2017 onwards, which offers significantly wider bandwidths–up to 200 MHz compared to the earlier 33 MHz. Since the signal-to-noise ratio scales with bandwidth ($\Delta\nu$) approximately as SNR $\propto \sqrt{\Delta\nu}$, the uGMRT observations result in $\sim 2.3\times$ increase in sensitive detections of both pulsars, improving the precision of timing and profile characterization.

Observations were conducted over a span exceeding ten years as part of long–term monitoring programs targeting two Fermi–detected millisecond pulsars (MSPs): J0248+4230 and J1207−5050. PSR J1207−5050 was previously undetected in band–3 (300–500 MHz) using the legacy GMRT and was therefore observed in band–4 (550–650 MHz) using the uGMRT, with a time resolution of 81.92 μs and a bandwidth of 200 MHz divided into 4096 frequency channels. In contrast, PSR J0248+4230, which exhibits a steep spectral index of $\alpha \sim -3.3$, remained detectable in band–3 and was thus observed using uGMRT band–3 with the same setup parameters. As earlier observations were conducted using the PA mode of the GMRT, both pulsars continued to be observed in PA mode with uGMRT which provides enhanced sensitivity but introduces intra-channel dispersion delays of approximately 0.3 ms and 0.007 ms for PSRs J0248+4230 and J1207−5050, respectively, due to their different observing frequencies and DMs.

Receiver Band	Frequency (MHz)	Mode[a]	T_{samp} (μs)
Band–3	300–500	PA	81.92
Band–4	550–750	PA	81.92
Band–5	1050–1450	PA	81.92

Table 1. Observation summary of the uGMRT receiver bands.

Followup (post–2017) observations were recorded with GMRT Wideband Backend (GWB) and each pulsar's observation typically lasted 30 to 45 minutes for over 50 sessions, providing a dense temporal sampling critical for maintaining phase connection and refining timing models. These multi–epoch observations ensured both a broad temporal baseline and sufficient data density for phase–connection. The raw voltage or filterbank data from each observing session were first inspected and processed for radio frequency interference (RFI), which is particularly prevalent at low observing frequencies (≤ 1 GHz). RFI excision was performed using the `gptool`[2] software, which enabled flagging of RFI contaminated frequency channels and time bins. In particular, gptool employed a Median Absolute Deviation (MAD)–based thresholding technique applied to the zero–dispersion measure (zero–DM) time series to identify and excise impulsive RFI. This step was critical for preserving the integrity of the data prior to further processing.

Following RFI mitigation, the data were folded into integrated pulse profiles using tools from the `PRESTO`[3] suite. Folding was performed using ephemerides from the initial discovery paper and updated iteratively. Folding involves aligning the incoming pulse signal with predicted pulse phases based on a provisional timing model, then averaging the signal over many rotations to build up a high S/N profile. This process sharpens the pulse shape, which is crucial for the next stage–template matching for time–of–arrival (ToA) extraction.

TOA calculation was performed by cross–correlating each folded profile with a high S/N standard template derived from the best observation of each pulsar. This was done using the Fourier–domain algorithm implemented using the tool of `TEMPO2`[4]. The TOA represents the time stamp at which a particular fiducial point (usually the peak or midpoint of the pulse) crosses the observer's location. The cross–correlation involves shifting and scaling the observed profile to match a high S/N standard template in the Fourier domain, using tools of `TEMPO2`. This process determines the phase offset between the template profile and observed profile, which is then converted into a TOA. The accuracy of this method depends on both the stability of the pulse shape and the fidelity of the template, making it essential for high–precision timing. Each TOA is accompanied by an uncertainty estimate, which reflects both instrumental noise and profile fidelity. These TOAs were then converted to the solar system barycenter using the Jet Propulsion Laboratory (JPL) DE436 ephemeris, correcting for observatory motion and clock delays.

The resulting TOAs were used to fit detailed timing models using `TEMPO2`. An iterative fitting process was

[2] GPTOOL: GitHub repository

[3] PRESTO: GitHub repository

[4] TEMPO2: GitHub repository

employed to derive phase–connected timing solutions. The timing model includes spin parameters (period P, period derivative $\dot{P}$), position (right ascension and declination), proper motion (μ_α, μ_δ), and dispersion measure (DM). In each iteration, model parameters were adjusted to minimize the timing residuals–the differences between observed TOAs and those predicted by the model. In both pulsars, this process revealed clear signatures of proper motion, which were subsequently included in the final models. The final residuals were examined for trends and noise characteristics. Diagnostic plots were generated with TEMPO2, including residuals versus time. In both cases, low root–mean–square (RMS) residuals were achieved, demonstrating the effectiveness of GMRT's phased-array system and the robustness of the data reduction pipeline.

This comprehensive observational and data analysis framework–spanning from raw data acquisition to precise TOA fitting–enabled the derivation of improved timing solutions for J0248+4230 and J1207−5050, which are detailed in the next section.

3. TIMING RESULTS

3.1. *Timing Study*

Pulsar timing is an iterative process wherein a timing model–comprising trial estimates of a pulsar's rotational and astrometric parameters–is fitted to time-of-arrival (TOA) data collected over multiple epochs. For both J0248+4230 and J1207−5050, we performed a least–squares fitting procedure using the software TEMPO2 pulsar timing package (Hobbs et al. 2006b). The initial timing models were adopted from the discovery paper by Bhattacharyya et al. (2021) and were gradually refined using our extended dataset spanning more than ten years.

We used a phase–connection approach to generate coherent timing solutions by fitting spin parameters (P, $\dot{P}$), sky position (RA, Dec), proper motion components (μ_α, μ_δ), dispersion measure (DM), and temporal evolution of DM($\dot{DM}$). For each pulsar, closely spaced TOAs helped maintain phase connection across all epochs. Cross–correlation of observed pulse profiles with high signal–to–noise standard templates allowed for sub–microsecond TOA precision, crucial for resolving subtle parameter variations. The fitting process aimed to minimize the timing residuals, defined as the difference between the observed and model–predicted TOAs.

The final timing solutions reveal significant improvements in astrometric precision compared to the initial models. Notably, we obtained the first measurements of proper motion for both pulsars, leading to inferred transverse velocities of 15.3 km s^{-1} for J0248+4230 and 43.9 km s^{-1} for J1207−5050. These velocities are consistent with the low–velocity population typically associated with recycled millisecond pulsars. The dispersion measures were also refined, and we detect no significant long–term DM variations over the baseline of our observations, consistent with a stable interstellar propagation path.

The residual plots generated using TEMPO2 may exhibit significant red noise or systematics. The root–mean–square (RMS) residuals were found to be low across all epochs, confirming the effectiveness of the PA–mode GMRT observations and the robustness of our TOA extraction and fitting procedures. For both pulsars, we find that the updated models accurately track pulse phase over the entire dataset, with residuals well within the timing precision expected for band–3 observations at 400 MHz.

No evidence of binary motion was detected for either pulsar, reaffirming their classification as isolated MSPs. Their clean timing behavior makes them excellent candidates for inclusion in future pulsar timing arrays (PTAs), particularly for enhancing sensitivity in sky regions currently underrepresented in PTA datasets.

In summary, our refined timing models for J0248+4230 and J1207−5050 significantly improve upon earlier ephemerides and provide precise astrometric and rotational parameters. These results are critical not only for characterizing the individual pulsars but also for evaluating their role in gravitational wave detection efforts and broader MSP population studies.

3.2. *Timing Residuals Results*

The initial timing analysis of PSR J0248+4230 was performed by Bhattacharyya et al. (2021) using legacy GMRT data spanning 2012–2017, covering 6 years of timing models. Timing parameters are sensitive to the length of the observational baseline: longer baselines allow more precise parameter estimation with reduced uncertainties and can reveal any long-term effects on the timing model. So we extended this timing baseline beyond 11 years by incorporating uGMRT observations from 2018 to 2025.

By including uGMRT data, which benefits from an enhanced bandwidth of 200 MHz, the uncertainties in time–of–arrival (ToA) measurements are reduced, leading to lower timing residuals. After this decadal-long timing study, the root–mean–square (rms) of the residuals was found to be 13.460 μs for PSR J0248+4230, as shown in Figure 2a. In this figure, black and blue points represent legacy GMRT band–3 and band–4 data, while the red points correspond to uGMRT band–3 observations. The plot includes ToAs with uncertainties limited to 15 μs, a threshold chosen to allow proper fitting of the pulsar's proper motion, which requires long-term data due to its annual periodicity.

For the first time, we have detected the transverse proper motion (μ_T) of PSR J0248+4230, measured to be 1.35 mas yr^{-1} (a $\sim 4\sigma$ detection). This transverse proper motion corresponds to a transverse velocity of 15.3 km s^{-1} (approximately 3.5σ significance), which is calculated using the distance YMW16, as shown in Table 2.

PSR Name	μ_T (mas/yr)	**Model**	d (kpc)	V_T (km/s)
J0248+4230	1.35 ± 0.35	YMW16	2.5	15.3 ± 4.4
J0248+4230	1.35 ± 0.35	NE2001	1.8	11.0 ± 3.2
J1207−5050	7.13 ± 0.47	YMW16	1.3	43.9 ± 5.3
J1207−5050	7.13 ± 0.47	NE2001	1.5	50.7 ± 6.0

Table 2. Proper motion and corresponding transverse velocities obtained using two Galactic electron density models (YMW16 and NE2001).

We also generated timing residuals using only seven years of uGMRT data (Figure 2b), applying the same 15 μs uncertainty cutoff for each ToA. This yields an rms of 9.464 μs. The uGMRT ToAs are more sensitive, which accounts for the reduced rms compared to the combined legacy and uGMRT dataset.

To further assess the impact of ToA precision, we produced timing residuals using only uGMRT data, this time with a stricter maximum ToA uncertainty of 5 μs (Figure 4a). This results in an rms of 4.329 μs, with all observations in band–3. Although these more sensitive ToAs (ΔToA ≤ 5 μs) reduce the number of usable data points compared to the 15 μs case, the improved precision leads to lower timing residuals and, consequently, more accurate and reliable timing model parameters. The updated parameter values are listed in the second column of Table 3. The initial pulsar localization was obtained through gated imaging, with the initial position taken from Bhattacharyya et al. (2021). With the updated timing analysis, the precise RA and Dec values are reported in Table 3. The updated period has a timing stability of $\propto 10^{-20}$, which means that the P value will change by 5 ns after 10^4 years of times scale.

For the isolated MSP J1207–5050, the initial timing analysis was carried out by Bhattacharyya et al. (2021) using only legacy GMRT observations spanning six years (2012–2017), which yielded an r.m.s. timing residual of 10.983 μs. In this work, we extend the temporal baseline beyond 2017 by incorporating uGMRT band–4 observations. This decadal extension results in an r.m.s. residual of 13.460 μs, when combining both legacy GMRT and uGMRT data, as shown in Figure 3a. Although the residuals increase compared to the legacy GMRT–only case, the extended baseline allows us to obtain more precise values of the model parameters (Table 3). By fitting for proper motion in this long-term study, we report the first detection of its value, 7.13 mas yr^{-1} (at $\sim 16\sigma$ significance). From this proper motion, we derived the transverse velocity using two Galactic electron-density models, as listed in Table 2.

We also reproduced the timing residuals using only uGMRT band–4 observations (Figure 3b), which yielded an r.m.s. residual of 9.729 μs — lower than that obtained using legacy GMRT data alone. For both legacy and uGMRT datasets, we initially used a maximum ToA uncertainty cut of 15 μs in order to probe long-term trends and to enable proper-motion fitting. To improve parameter precision, we then restricted the ToA uncertainties of uGMRT observations to below 5 μs, producing fewer ToAs per epoch. This yielded an r.m.s. residual of 3.159 μs (Figure 4), and correspondingly more precise model parameters (Table 3).

It is important to note that the upgrade from the legacy GMRT to the uGMRT introduced a constant time offset of 1.34217728 μs between the respective ToAs. This offset was accounted for by including a constant phase jump in the timing model prior to the analysis.

4. MULTI–WAVELENGTH STUDY

4.1. *Gamma–Ray Folded Profiles:*

In this section, we assess the gamma–ray pulse profiles and spectral energy distributions (SEDs) of PSRs J1207−5050 and J0248+4230 using updated timing models and LAT data products.

4.2. *PSR J1207−5050: LAT Timing and Gamma–ray Profile Evolution*

We compared gamma–ray foldings of PSR J1207−5050 using LAT timing parameters from the Third *Fermi* Pulsar Catalog (3PC) with those derived from our extended decadal radio timing campaign. Notably, the LAT–based timing model yields a higher test statistic ($TS_{\rm H} = 303.87$) compared to the folded gamma–ray profile folded with radio–based timing model ($TS_{\rm H} = 249.69$), and shows narrower, more pronounced peaks in the gamma–ray profile (Figure 5b).

Figure 5b displays the cumulative H–test statistic computed by integrating backward in time. Both the LAT and radio solutions indicate signal weakening or loss before MJD 56500. Although the LAT profile appears cleaner, it may be subject to bias due to gamma–

Table 3. Pulsar Parameters from Radio Timing and Derived Parameters

Name	**J0248+4230**	**J1207–5050**
Gated imaging position*		
Right ascension (J2000)	$02^h48^m29^s(7)$	$12^h07^m21^s(5)$
Declination (J2000)	+42°30'13"(4)	50°50'30"(10)
Parameters from radio timing*		
Right ascension (J2000)	$02^h48^m31^s.0303(1)$	$12^h07^m22^s.40392(7)$
Declination (J2000)	+42°30'20".49(5)	50°50'38".680(1)
Proper motion in RA (mas yr^{-1})	0.6(6)	1.4(6)
Proper motion in DEC (mas yr^{-1})	1.4(5)	0.1(6)
Frequency f (Hz)	384.49193525267(4)	206.49339501709(3)
Frequency derivative $\dot{f}$ (Hz s^{-1})	$2.944(7)\times10^{-15}$	$2.586(2)\times10^{-15}$
Period epoch (MJD)	56588.0	56563.0
Dispersion measure DM (pc cm^{-3})	48.26341(3)	50.67
DM 1^{st} derivative DM1	–	–
Timing Data Span	56375.3–62084.8	55728.6–62803.8
Number of TOAs	420	519
Reduced χ^2	1.07	1.06
Post–fit residual rms (ms)	0.0056	0.0056
Derived parameters		
Period (ms)	2.600837485636164(3)	4.842757232802074(2)
Period Derivative ($\dot{P}$) (s/s)	$1.687(3)\times10^{-20}$	$6.067(7)\times10^{-20}$
Total time span (yr)	4.6	5.6
Energy loss $\dot{E}$ (erg/s)	3.8×10^{34}	6.4×10^{33}
Characteristic age (yr)	1.6×10^{9}	1.3×10^{9}
Surface magnetic field (Gauss)	2.1×10^{8}	5.4×10^{8}
Shklovskii distance (kpc)†	2.5	1.8
DM distance (kpc)††	3.5	2.0

ray signal modeling and background subtraction. In contrast, the radio–derived solution is free from such biases and thus preferable for detailed profile morphology studies. Nevertheless, from a timing model perspective, the LAT solution may be closer to the underlying rotational ephemeris.

4.3. *Spectral Properties and Energy Distribution*

The SED of PSR J1207−5050 presents an excellent example of curvature radiation from a narrowly distributed electron population. Two SED models were fitted: the default Log Parabola from the 4FGL catalog, and the PLEC4 (Power–Law with Exponential Cutoff) model preferred for MSPs.

The fitted PLEC4 parameters are:
$n_0 = 4.88 \times 10^{-13}$,
$\gamma = 1.83$,
$d = 1.12$,
$b = 2/3$,
$E_0 = 1.59$ GeV.

Following Equation 18 of 3PC, the curvature parameter $d_p = 1.24$, which is remarkably close to the theoretical prediction $d_p = 4/3$ for curvature radiation by monoenergetic electrons. This match supports interpretations discussed in 3PC Section 6, where MSPs with $\dot{E} \sim 2.1 \times 10^{33}$ erg/s (just above the gamma–ray deathline) exhibit SEDs sharply peaked near 2 GeV, consistent with curvature emission limits.

4.4. *PSR J0248+4230: Gamma–ray Evolution*

For PSR J0248+4230, Figure 5a shows the LAT gamma–ray folded profile using the updated timing solution. The profile displays broadened components and energy–dependent structures, though with lower $TS_{\rm H}$ compared to J1207−5050. The data reinforce this pulsar's lower gamma–ray flux and signal significance, consistent with its higher DM and potential distance.
These multi–wavelength analyses, combining long–term radio timing and *Fermi*–LAT data, yield deeper insights into MSP energetics, emission geometry, and magnetospheric particle dynamics.

4.5. *Optical Surveys*

We also searched for optical counterparts of the MSPs J1207−5050 and J0248+4230 in optical and near-infrared surveys. The deepest data available were from SkyMapper (Wolf et al. 2018; Onken et al. 2019) in the optical band. However, no sources are detected within sub-arcsecond radii of the given RA and Dec positions derived from the updated radio timing model.

5. POLARIZATION STUDY

For **PSR J1207−5050**, the polarization profiles are presented in Figure 1a. The white, red, and blue curves represent the total intensity, linear, and circular polarization, respectively, plotted using `psrpype`. The integrated profile in the frequency range 0.7–1.4 GHz shows a very faint detection of J1207−5050, plotted using `psrchive`. The results suggest the following:

- A positive spectral index ($\alpha > 0.43$).
- A possible spectral turnover above ~ 650 MHz.
- Some evidence of noise or RFI contamination, indicating that further investigation or new polarization observations are required.

From the Parkes UWL observations, we carried out a polarization study of PSR J1207−5050, as illustrated in Figure 9. The polarization analysis shows that the position angle (PA) sweep across the on-pulse window remains nearly flat, with only minor variations consistent with measurement uncertainties. The emission is found to be strongly linearly polarized, with a fractional linear polarization of ∼93.25%, and no significant circularly polarized component is detected. This indicates that the magnetospheric emission of PSR J1207−5050 is highly ordered, with little evidence of depolarization due to propagation effects or orthogonal mode mixing. Such characteristics make this pulsar a useful probe for investigating the geometry of the emission region and the role of orthogonal polarization modes in millisecond pulsars. Furthermore, the Parkes UWL data show that the integrated pulse profile is not detectable beyond ∼1.35,GHz, providing an observational constraint on the radio emission bandwidth.

For **PSR J0248+4230**, a polarization study has not been performed yet. This is because its right ascension (RA) and declination (Dec) position is not visible to the Parkes Radio Telescope, making polarization observations infeasible at present.

6. COMPARISON WITH PTA–LISTED MSPS

Pulsar Timing Arrays (PTAs) such as the European Pulsar Timing Array (EPTA), North American Nanohertz Observatory for Gravitational Waves (NANOGrav), and Parkes Pulsar Timing Array (PPTA) form a global collaboration under the International Pulsar Timing Array (IPTA) to detect low–frequency gravitational waves. These efforts rely on high–precision timing of millisecond pulsars (MSPs) with low red noise, stable profiles, and broad sky coverage. Pulsars included in PTAs are carefully selected based on their rotational stability, low timing noise, and strategic placement in the sky.

To assess the eligibility of PSRs J0248+4230 and J1207−5050 for inclusion in PTA projects, we compared several of their timing and positional parameters with those of the 42 EPTA–listed pulsars. The comparison included spin period (P_0), dispersion measure (DM), galactic coordinates (l, b), and timing residuals.

Figure 8 presents key comparisons:

- **Dispersion Measure (DM) vs Spin Period ($\mathbf{P_0}$):** Both J0248+4230 (DM $\sim$ 48.3 pc cm^{-3}, $P_0 \sim 2.60$ ms) and J1207−5050 (DM $\sim$ 50.6 pc cm^{-3}, $P_0 \sim 4.84$ ms) lie within the core parameter space occupied by EPTA MSPs. This confirms their suitability for high–precision timing, as moderate DM values reduce ISM scattering effects and aid in timing model stability.

- **Galactic Latitude vs Spin Period:** PTA collaborations seek to optimize sky coverage to reduce timing correlations. J0248+4230, located at a galactic latitude of −15.32°, and J1207−5050, at +11.42°, complement existing PTA spatial distributions. Their inclusion would improve sky sampling, particularly in less–dense regions of current PTA networks.

- **Elliptical Latitude Distribution:** A comparison in elliptical coordinate space shows that both MSPs occupy regions that enhance hemispheric balance, which is critical for reducing spatial covariance in PTA datasets and improving gravitational wave background detection sensitivity.

At 400 MHz, PSR J0248+4230 has a flux density of $S_{400} = 7.5$ mJy. This places it slightly above the median flux density of available ATNF–listed MSPs (~ 6.8 mJy), and close to the median value of the NANOGrav sample (~ 7.05 mJy). Although the median flux density of EPTA MSPs is somewhat higher (~ 11 mJy), a significant number of pulsars in both the EPTA (29%) and NANOGrav (53%) samples have flux densities comparable to or below that of J0248+4230. This indicates that its flux density is sufficient for inclusion in PTA efforts, especially when combined with its favorable angular separation and timing stability.

Another critical requirement for including a new pulsar in a PTA is that it should sample a different region of the sky. This spatial diversity improves the strength of the correlation in the so-called *Hellings and Downs* curve, which is essential for gravitational wave detection. From the angular separation plots (Figure 7), it is evident that both J0248+4230 and J1207−5050 are well angularly separated from all pulsars listed in both the EPTA and NANOGrav catalogs. The angular distance statistics are as follows:

- **EPTA Angular Distance Statistics:**
  - From J1207−5050 → Mean: 93.31°, Median: 89.98°
  - From J0248+4230 → Mean: 99.13°, Median: 104.43°
- **NANOGrav Angular Distance Statistics:**
  - From J1207−5050 → Mean: 105.09°, Median: 110.81°
  - From J0248+4230 → Mean: 89.66°, Median: 91.61°

These results confirm that both pulsars are positioned at sufficiently large angular distances from existing PTA pulsars, making them excellent candidates for improving the overall sky coverage and sensitivity of PTA experiments.

These comparisons affirm that J0248+4230 and J1207−5050 meet many of the key criteria used by PTAs for pulsar selection. Their favorable timing characteristics and sky position make them valuable additions to future PTA efforts, potentially enhancing gravitational wave sensitivity and sky coverage.

7. SINGLE-PULSE AND BRIGHT-PULSE ANALYSIS

In addition to integrated profile studies, we investigated the single-pulse emission behavior of the two MSPs. Single-pulse studies provide important insights into emission mechanisms, mode changing, and pulse-to-pulse variability, which can affect the timing precision of PTA pulsars.

For PSR J0248+4230, we report the detection of individual single pulses above the noise level in uGMRT band–3 observations. Figure **??** shows two representative single-pulse detections compared with the integrated profile. The first single pulse peaks at phase ~ 0.67, coincident with the integrated profile peak, while the second bright pulse peaks at phase ~ 0.80, slightly offset from the average profile. These pulses demonstrate significant pulse-to-pulse variability, with some individual pulses being several times stronger than the mean flux density of the integrated profile.

No clear evidence of single-pulse detections was found for PSR J1207−5050, likely due to its comparatively lower flux density and weaker integrated profile at the observed frequencies.

The presence of detectable single pulses from J0248+4230 not only provides an additional handle to study its emission physics but also reinforces its utility in PTA applications. Bright pulses can be used to obtain high signal-to-noise timing measurements, thereby improving the robustness of ToA extraction in noisy or RFI-affected datasets.

8. DISCUSSION AND CONCLUSIONS

In this work, we have presented a decade-long timing study of the two GMRT-discovered isolated millisecond pulsars J0248+4230 and J1207−5050, significantly extending the temporal baselines used in earlier studies. Using the upgraded sensitivity of the uGMRT and a carefully constructed pipeline for TOA extraction and RFI mitigation, we have refined their rotational and astrometric parameters to unprecedented precision.

A major outcome of this study is the *first detection of proper motion for both pulsars*, corresponding to transverse velocities of $\sim$15 km s^{-1} for J0248+4230 and $\sim$44 km s^{-1} for J1207−5050. These velocities are well within the distribution expected for recycled MSPs, supporting the view that they belong to the low-kick, stable velocity population associated with binary evolution and recycling. Their measured stability (with timing residual rms values of just a few microseconds when using the most precise uGMRT TOAs) further highlights their potential as long-term precision clocks.

The gamma-ray profiles derived using updated timing ephemerides confirm the detection of both pulsars as Fermi-LAT emitters. For J1207−5050, the phase-aligned gamma-ray profile shows strong, sharp peaks consistent with curvature radiation, while J0248+4230 displays a broader profile with reduced significance, possibly linked to distance and line-of-sight effects. The spectral energy distribution of J1207−5050, in particular, closely follows theoretical expectations for curvature radiation from MSP magnetospheres. The absence of optical counterparts in SkyMapper data further reinforces their classification as isolated pulsars.

In the context of pulsar timing arrays (PTAs), both MSPs compare favorably with current EPTA-listed sources in terms of dispersion measure, spin period, and timing precision. Their intermediate Galactic latitudes provide complementary sky coverage to existing PTA members, and their demonstrated rotational stability suggests that they could contribute meaningfully to the global PTA effort aimed at detecting nanohertz gravitational waves.

In conclusion, this study establishes J0248+4230 and J1207−5050 as stable, well-characterized isolated millisecond pulsars. By extending their timing baselines to more than a decade and detecting their proper motions for the first time, we have substantially reduced parameter uncertainties and strengthened their scientific value. Continued long–term monitoring with the uGMRT and other facilities will not only refine their timing further but also test their suitability as PTA candidates, thereby enhancing the sensitivity of gravitational-wave searches and enriching our understanding of the recycled pulsar population.

REFERENCES

Agazie, G., Anumarlapudi, A., Archibald, A. M., et al. 2023, ApJL, 951, L8, doi: 10.3847/2041-8213/acdac6

Alpar, M. A., Cheng, A. F., Ruderman, M. A., & Shaham, J. 1982, Nature, 300, 728, doi: 10.1038/300728a0

Antoniadis, J., Arumugam, P., Arumugam, S., et al. 2023. https://arxiv.org/abs/2306.16214

Bhattacharya, D. 1991, Phys. Rep., 203, 1, doi: 10.1016/0370-1573(91)90064-s

Bhattacharyya, B., Roy, J., Johnson, T. J., et al. 2021, ApJ, 910, 160, doi: 10.3847/1538-4357/abe4d5

Gupta, Y., Ajithkumar, B., Kale, H. S., et al. 2017, Current Science, 113, 707, doi: 10.18520/cs/v113/i04/707-714

Hewish, A., Bell, S. J., Pilkington, J. D. H., Scott, P. F., & Collins, R. A. 1968, Nature, 217, 709, doi: 10.1038/217709a0

Hobbs, G., Archibald, A., Arzoumanian, Z., et al. 2010

Hobbs, G. B., Edwards, R. T., & Manchester, R. N. 2006a, MNRAS, 369, 655, doi: 10.1111/j.1365-2966.2006.10302.x

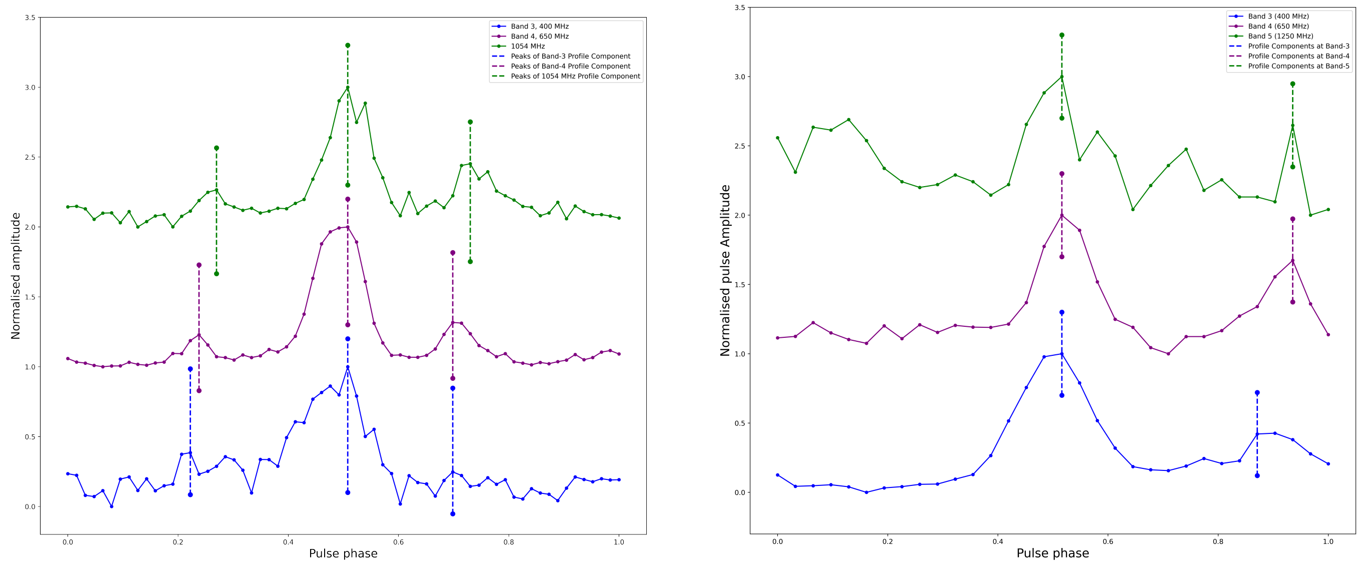
—. 2006b, MNRAS, 369, 655, doi: 10.1111/j.1365-2966.2006.10302.x

Lorimer, D. R., & Kramer, M. 2004, hpa, 4

Onken, C. A., Wolf, C., Bessell, M. S., et al. 2019, PASA, 36, e033, doi: 10.1017/pasa.2019.27

Swarup, G. 1991, in Astronomical Society of the Pacific Conference Series, Vol. 19, IAU Colloquium 131: Radio Interferometry. Theory, Techniques, and Applications, ed. T. J. Cornwell & R. A. Perley, 376–380

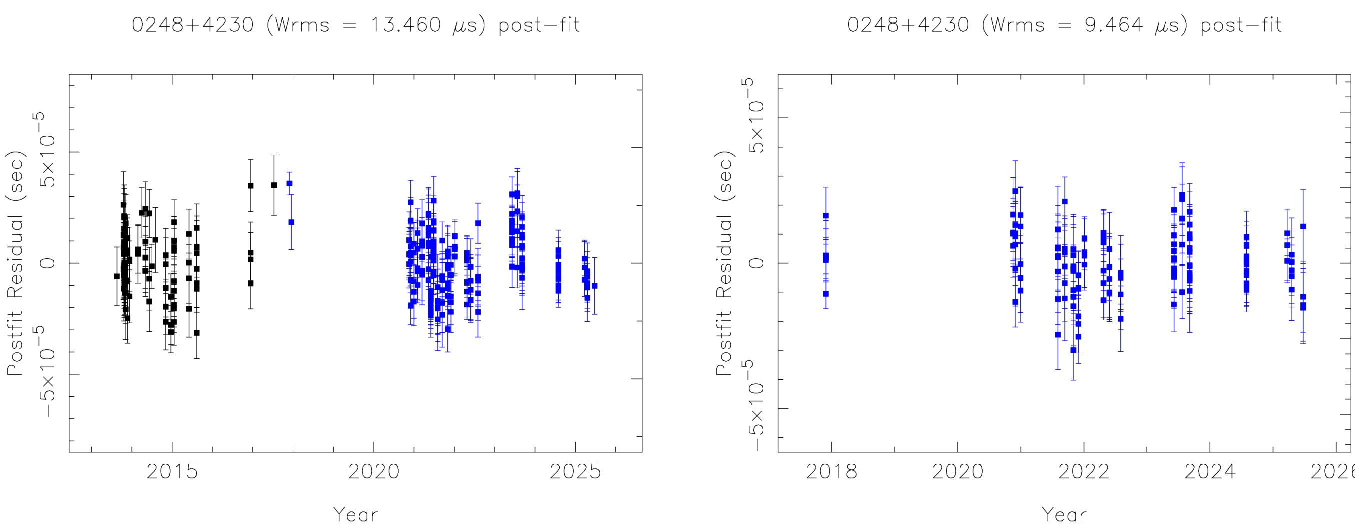
Wolf, C., Onken, C. A., Luvaul, L. C., et al. 2018, PASA, 35, e010, doi: 10.1017/pasa.2018.5



(a) Profile evolution of PSR J1207−5050 at band–3 (blue) and band–4 (purple) and the green curve shows the Parkes UWL profile observed at a central frequency of 1.02 GHz (0.7–1.35 GHz). The corresponding dashed vertical lines mark the peak positions of the 1st, 2nd, and 3rd components at band–3, band–4, and 1.02 GHz, respectively.

(b) Profile evolution of PSR J0248−4230 at band–3 (purple), band–4 (green,) and band–5 (orange). The dashed vertical lines (purple, green, and orange) mark the peak positions of the the 1st and 2nd components of the at band–3, band–4, and band–5 profiles, respectively.

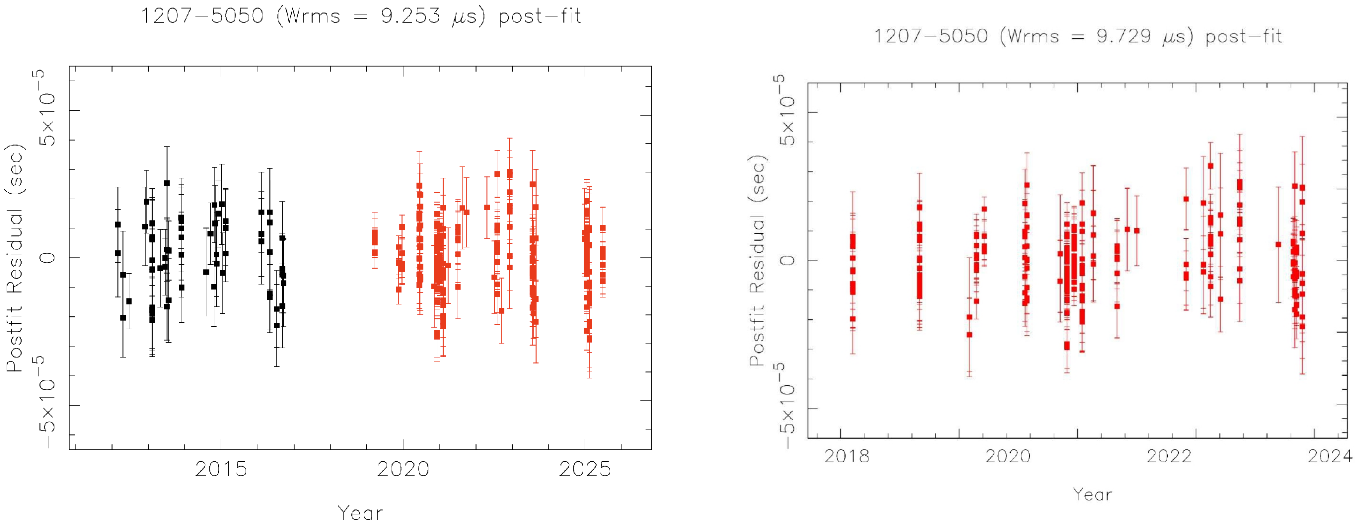
Figure 1. Profile evolution for PSR J0248+4230 and J1207−5050 at band–3, band–4 and band–5.



(a) Postfit timing residual from 12 years of timing for PSR J0248+4230 using legacy GMRT and uGMRT observations. Black and Red are data points correspond to band–3 and band–4 observations using legacy GMRT. Blue data points corresponds to band–3 uGMRT observations.

(b) Postfit timing residual from 7 years of timing for PSR J0248+4230 using uGMRT observations. Blue data points corresponds to band–3 uGMRT observations.

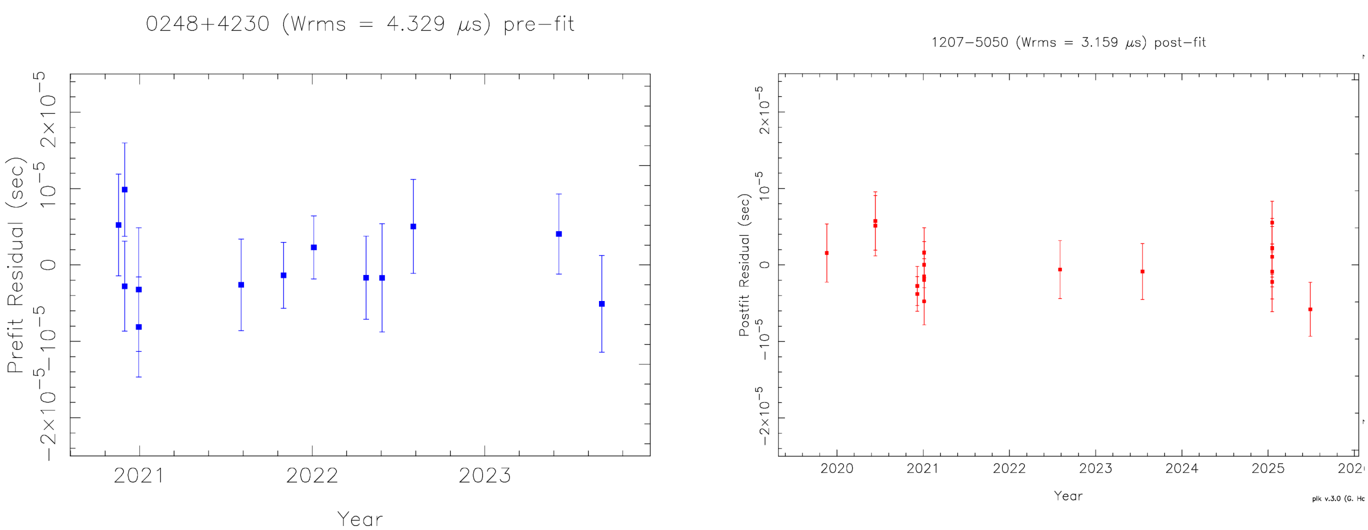
Figure 2. Timing residual plots for PSR J0248+4230 as a function of time.



(a) Postfit timing residual from 12 years of timing for PSR J1207−5050 using legacy GMRT and uGMRT observations. Black and Red are data points correspond to band–4 observations using legacy GMRT and uGMRT.

(b) Postfit timing residual from 6 years of timing for PSR J1207−5050 using uGMRT observations. Red data points correspond to band–4 uGMRT observations.

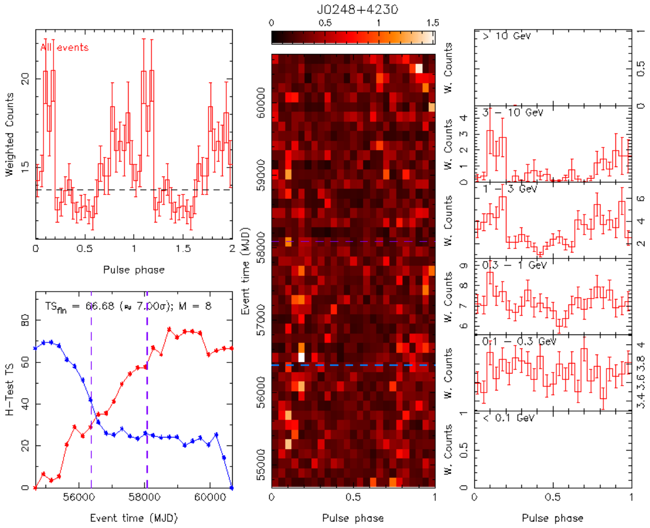
Figure 3. Timing residual plots for PSR J1207−5050 as a function of time.



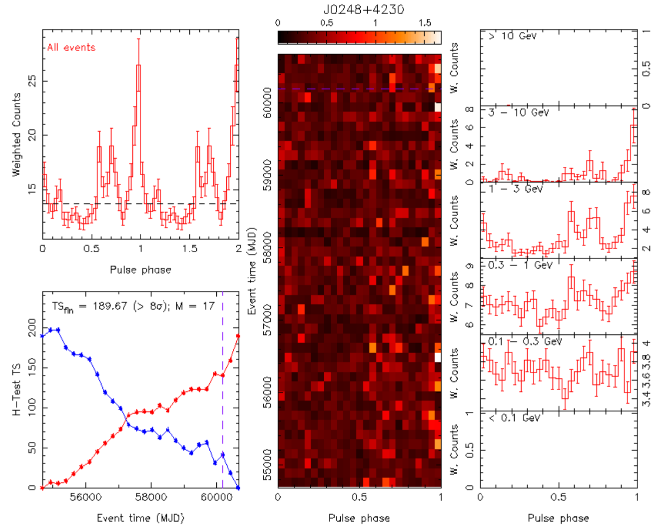
(a) Postfit timing residual from 6 years of timing for PSR J1207−5050 using uGMRT observations. Blue data points corresponds to band–3 uGMRT observations.

(b) Postfit timing residual from 4 years of timing for PSR J1207−5050 using uGMRT observations. Red data points corresponds to band–4 uGMRT observations.

Figure 4. Timing residuals for PSR J0248+4230 and J1207−5050 as a function of time with ToAs uncertainty upto 5 μs

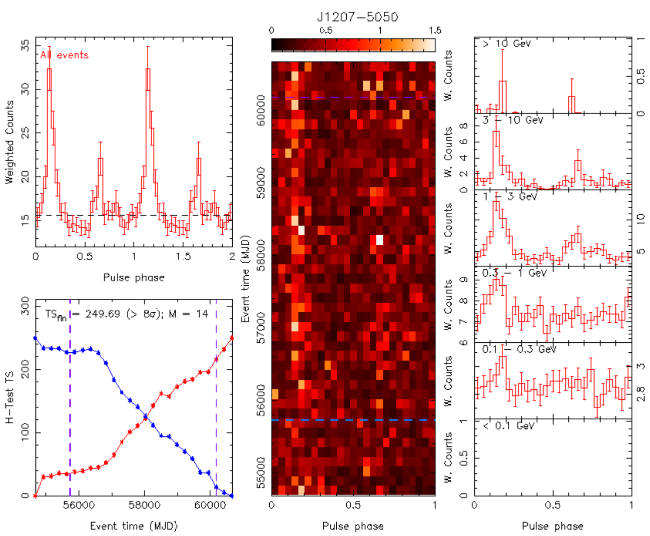


(a) Fermi LAT Gamma–ray folded profile of PSR J1207−5050, using initial timing parameters.

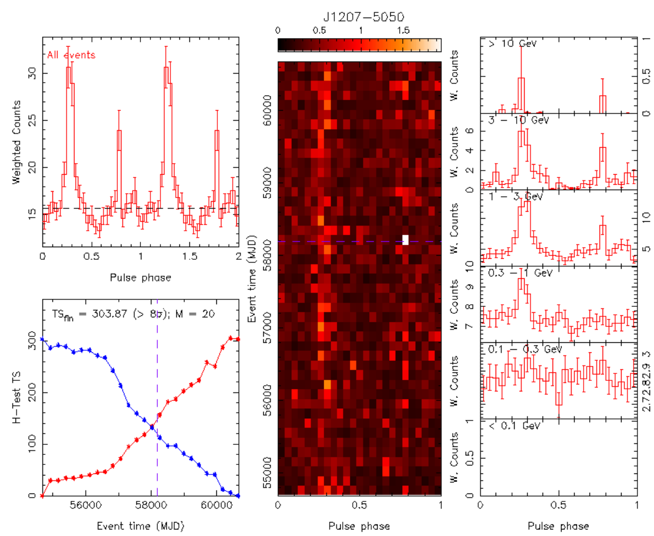


(b) Fermi LAT Gamma–ray folded profile of PSR J1207−5050, using timing parameters from after the decadal extension.

Figure 5. Fermi LAT Gamma–ray folded profile of PSR J1207−5050.

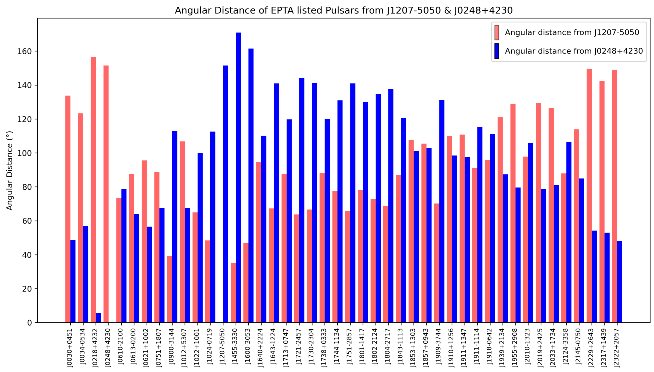


(a) Fermi LAT Gamma–ray folded profile of PSR J0248+4230, using initial timing parameters.

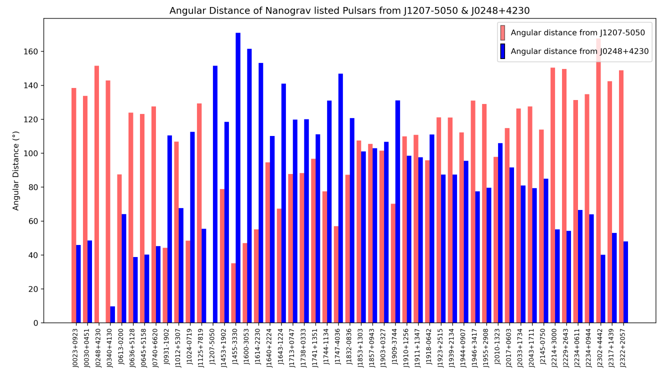


(b) Fermi LAT Gamma–ray folded profile of PSR J0248+4230, using timing parameters from after the decadal extension.

Figure 6. Fermi LAT Gamma–ray folded profiles of PSR J1207−5050.

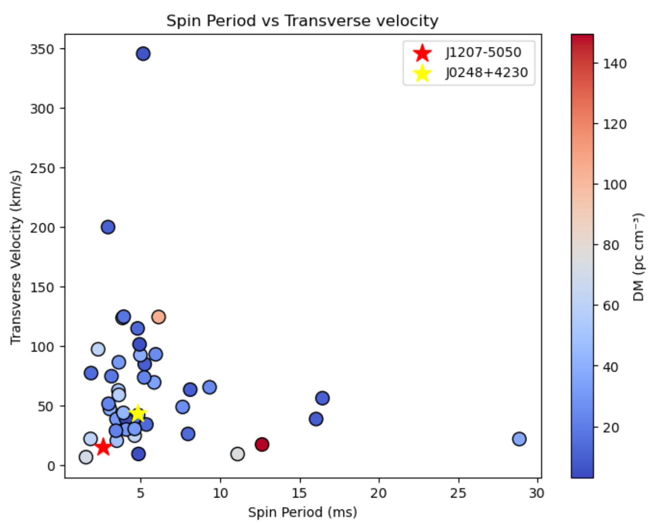


(a) Angular distance of PSR J1207−5050 (red color) and J0248+4230 (blue color) from the EPTA listed pulsar .

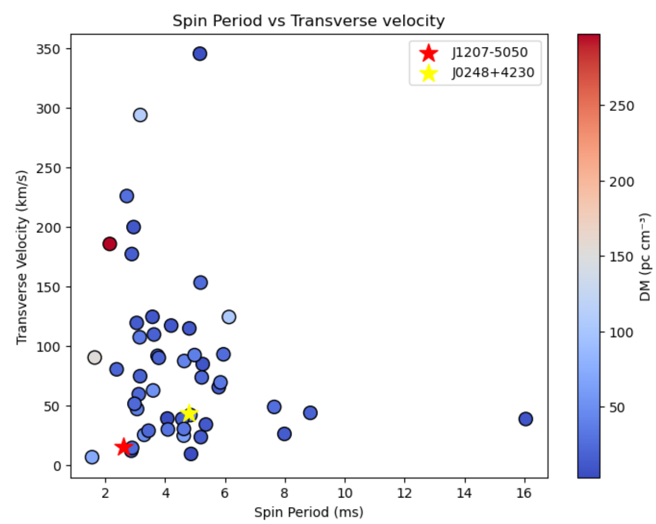


(b) Angular distance of PSR J1207−5050 (red color) and J0248+4230 (blue color) from the NANOGrav listed pulsar .

Figure 7. Angular distance of PSR J1207−5050 (red color) and J0248+4230 (blue color) from EPTA and NANOGrav listed pulsars.



(a) Transverse velocity (km/s) vs. spin period (ms) for pulsars in the EPTA catalog. Pulsars J0248–4230 and J1207–5050 are highlighted in red and blue, respectively.



(b) Transverse velocity (km/s) vs. spin period (ms) for pulsars in the NANOGrav catalog. Pulsars J0248–4230 and J1207–5050 are highlighted in yellow and red respectively.

Figure 8. Comparison of transverse velocity and spin period for PTA-listed pulsars from the EPTA and NANOGrav catalogs, with J0248–4230 and J1207–5050 highlighted in yellow and red respectively.

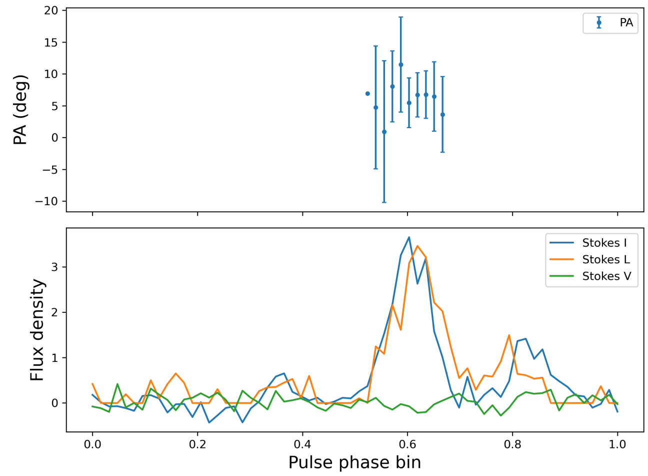


Figure 9. Polarization profile of PSR J1207−5050. **Top:** Position angle (PA) of the linearly polarized emission, plotted only in the on-pulse phase window, with error bars. **Bottom:** Pulse profile showing Stokes I (total intensity), Stokes L (debiased linear polarization), and Stokes V (circular polarization) as a function of pulse phase.